

Lecture

Agenda

- Scaling
- Labelling

Reference

- Dufour & Meeks (2024) Ch 3

D3 data workflow

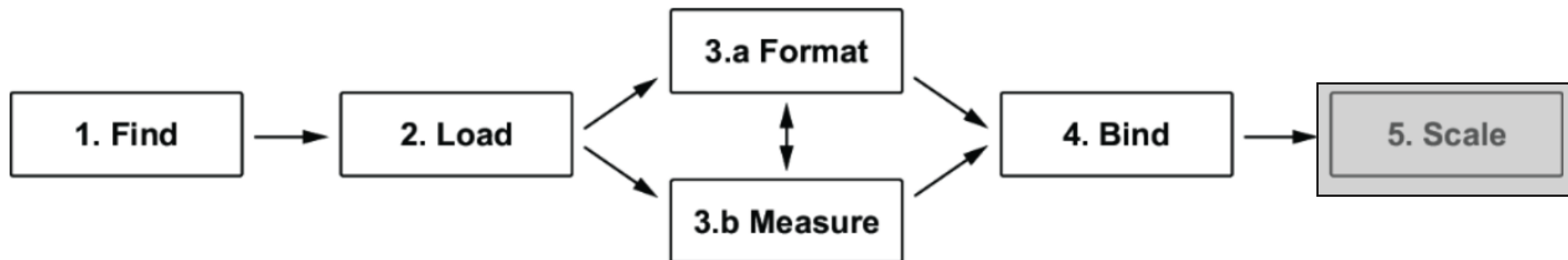


Figure 3.2 Simplified diagram of the D3 data workflow

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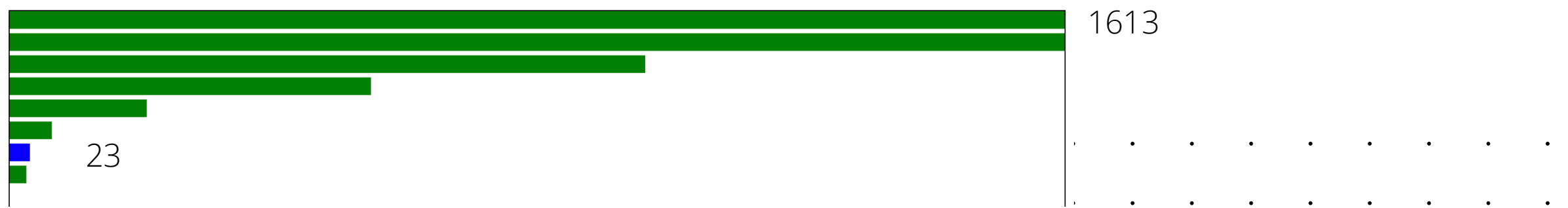
Scaling

"Modified land" is MUCH higher count than "rainforest and related scrub"

Modified land is 'off' the visible svg

Scaling values to available space will help with this problem

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Scaling the chart

Doman values are mapped proportionally to the range of pixels available in **svg**.

Different scales for different data types

- d3.scaleLinear() - for continuous data
- d3.scaleBand() - for discrete (qualitative data)

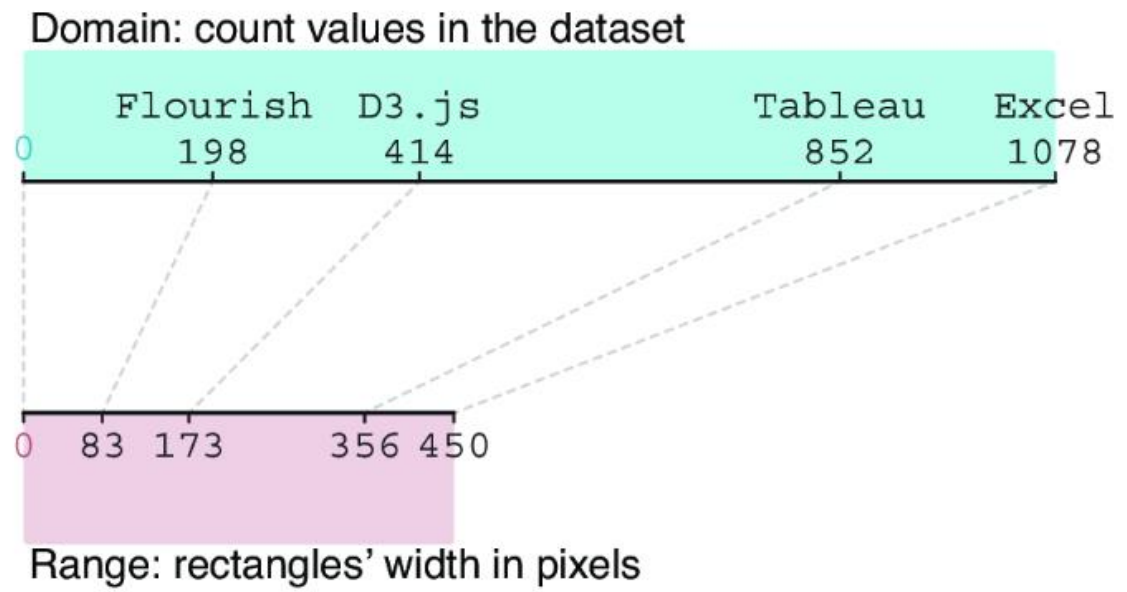
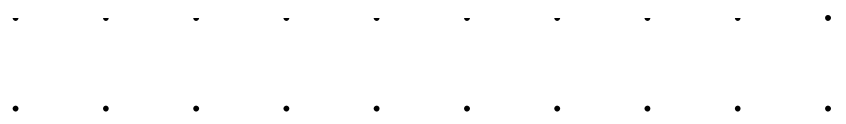


Figure 3.26 Count values from the data mapped by the linear scale into the bars' width



Linear Scale

```
const xScale = d3.scaleLinear()
  .domain([0, 1620])
  .range([0, 1000]);
```

...

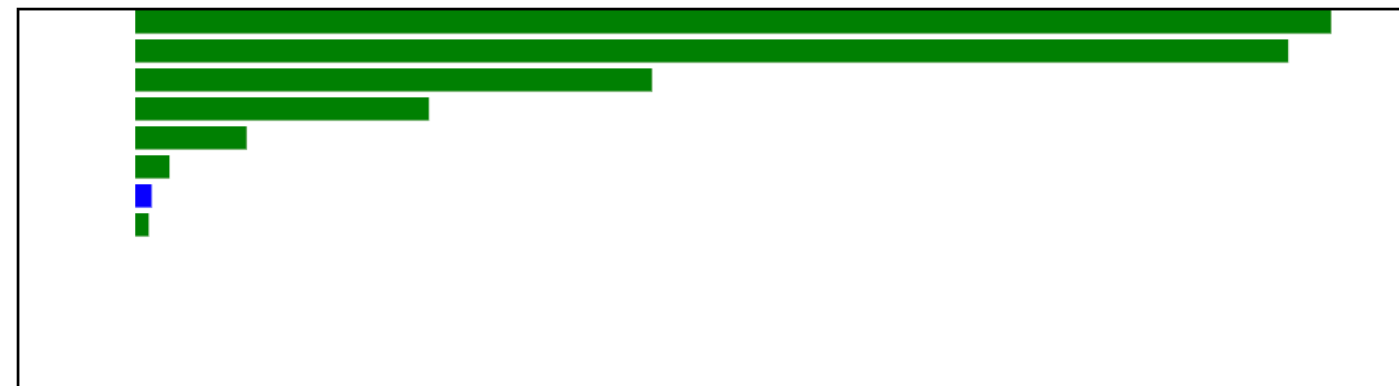
```
.attr("width", d => xScale(d.sighting))
```

...

```
.attr("x", 100)
```

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100px



Was

```
.attr("width", d => d.sighting)
```

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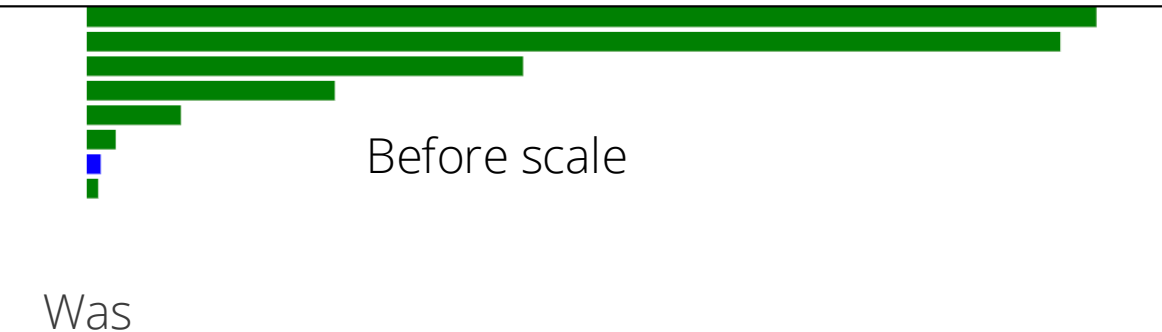
Band Scale

```
const yScale = d3.scaleBand()
  .domain(data.map(d => d.habitat))
  .range([0, 500])

...

.attr("y", d => yScale(d.habitat))
```

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Was

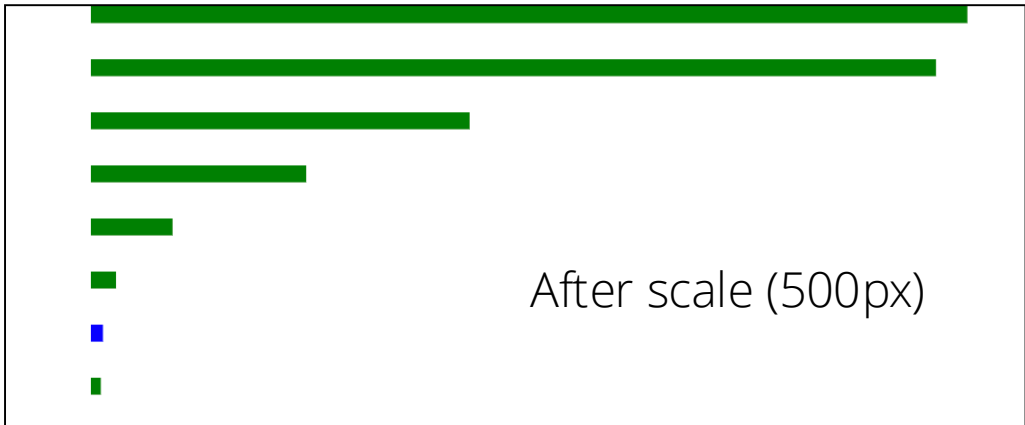
```
.attr("y", (d, i) => (barHeight + 5) * i)
```

```
[
  { habitat: "Rainforest", sighting: 120 },
  { habitat: "Desert", sighting: 80 },
  { habitat: "Grassland", sighting: 150 }
]
```

```
["Rainforest", "Desert", "Grassland"]
```

8 habitats, equally spread over svg height

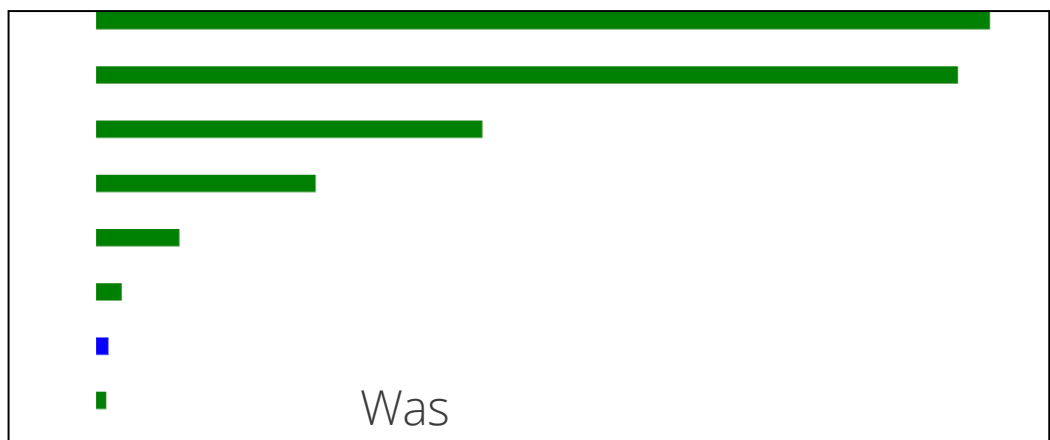
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Band Scale

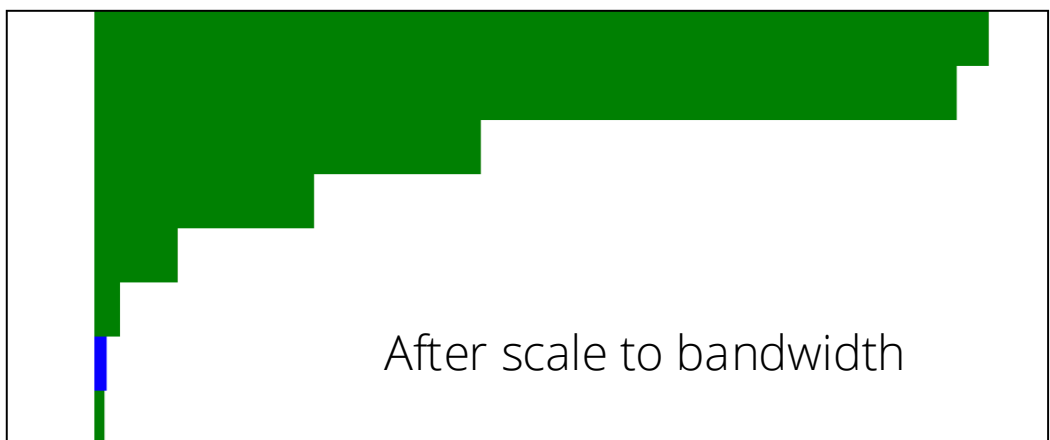
```
const yScale = d3.scaleBand()  
  .domain(data.map(d => d.habitat))  
  .range([0, 500])  
  
...  
  
  .attr("height", yScale.bandwidth())
```

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Was
`.attr("height", barHeight)`

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After scale to bandwidth

Band Scale

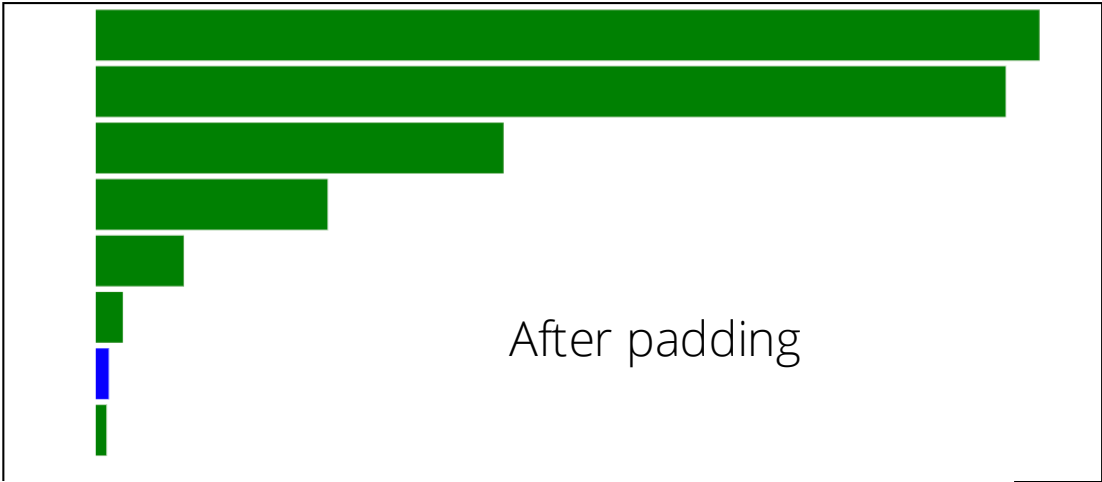
```
const yScale = d3.scaleBand()  
  .domain(data.map(d => d.habitat))  
  .range([0, 500])  
  .padding(0.1);
```

...

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Band scale

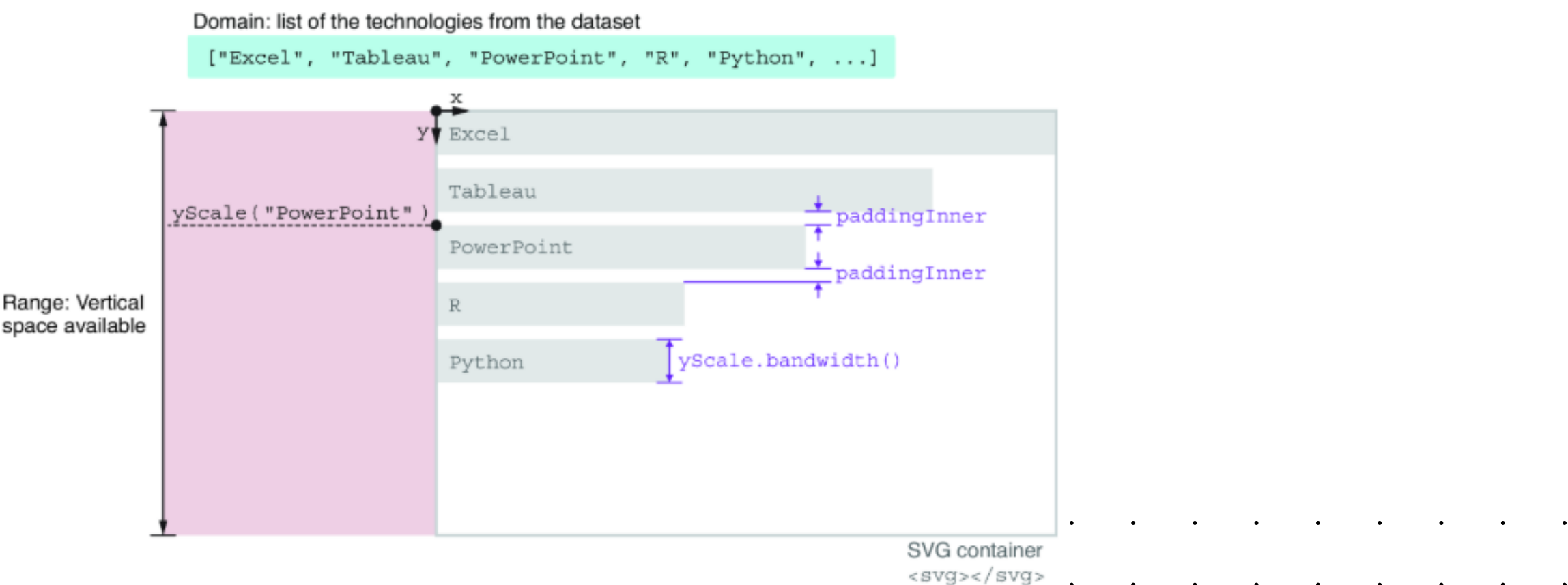


Figure 3.29 How the band scale distributes the list of technologies within the available vertical space

Adding Labels

Bind rectangles and angles together in a group

```
const barAndLabel = svg
  .selectAll("g")
  .data(data)
  .join("g")
```

Move rectangles and labels together

```
.attr("transform", d => `translate(0, ${yScale(d.habitat)})`);
```

Makes a group that starts at y = 0

```
<svg viewBox="0 0 1200 1600" style="border: 1px solid black;">
  <g transform="translate(0, 6.1728395061728065)"></g>
  <g transform="translate(0, 67.90123456790121)"></g>
  <g transform="translate(0, 129.62962962962962)"></g>
  <g transform="translate(0, 191.358024691358)"></g>
  <g transform="translate(0, 253.0864197530864)"></g>
  <g transform="translate(0, 314.8148148148148)"></g>
  <g transform="translate(0, 376.5432098765432)"></g>
  <g transform="translate(0, 438.2716049382716)"></g>
</svg>
```

```

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```

Adding Labels

Organisation of groups

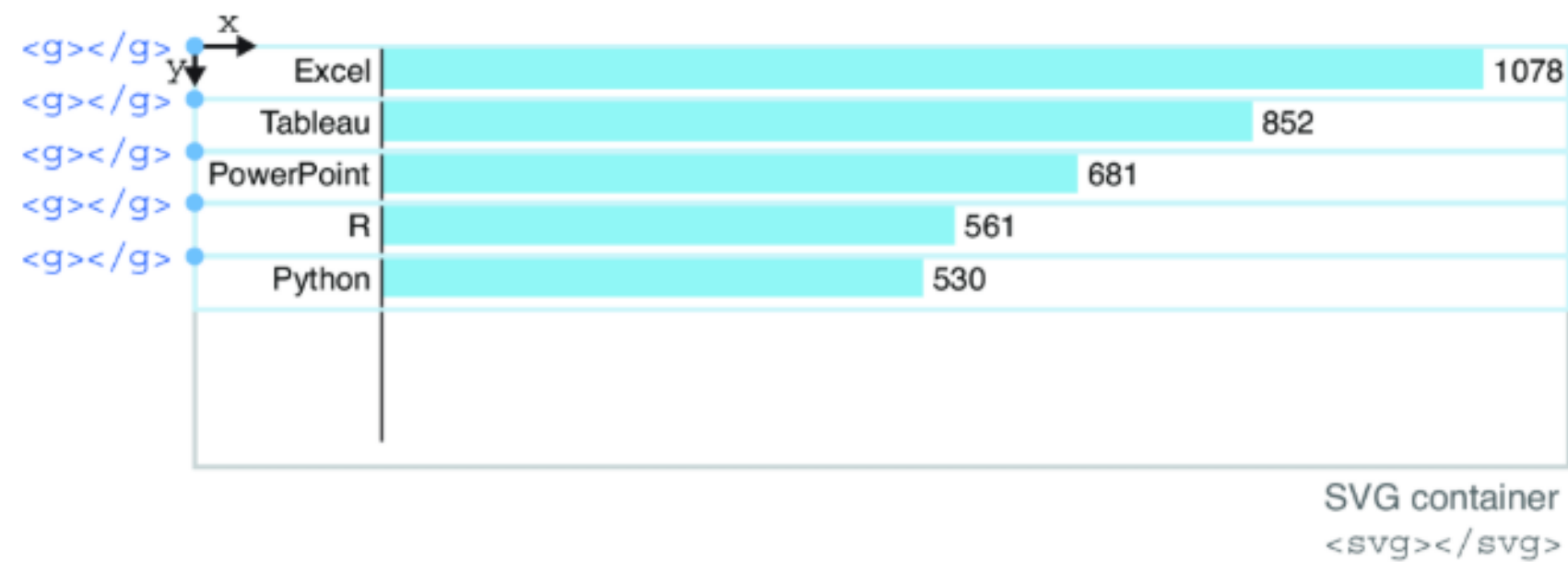


Figure 3.30 Groups positioned within the SVG container and encapsulating their descendent rectangle and labels

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Add rectangles to the group

barAndLabel

```
.append("rect");
```

```

<svg viewBox="0 0 1200 1600" style="border: 1px solid black;">
  <g transform="translate(0, 6.1728395061728065)">
    <rect></rect>
  </g>
  <g transform="translate(0, 67.90123456790121)"> ... </g>
  <g transform="translate(0, 129.62962962962962)"> ... </g>
  <g transform="translate(0, 191.35822460135822)"> ... </g>

```

```

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```

Add rectangles to the group

barAndLabel

```

.append("rect")
  .attr("class", d => {
    console.log(d);
    return `bar bar-${d.sighting}`;
  })
  .attr("width", d => xScale(d.sighting))
  .attr("height", yScale.bandwidth())
  .attr("x", 100)
  .attr("y", 0)
  .attr("fill", "green")
  .attr("fill", d => d.habitat === "Rainforest and related scrub" ? "blue" : "green");
};

```

.attr("y", d => yScale(d.habitat))

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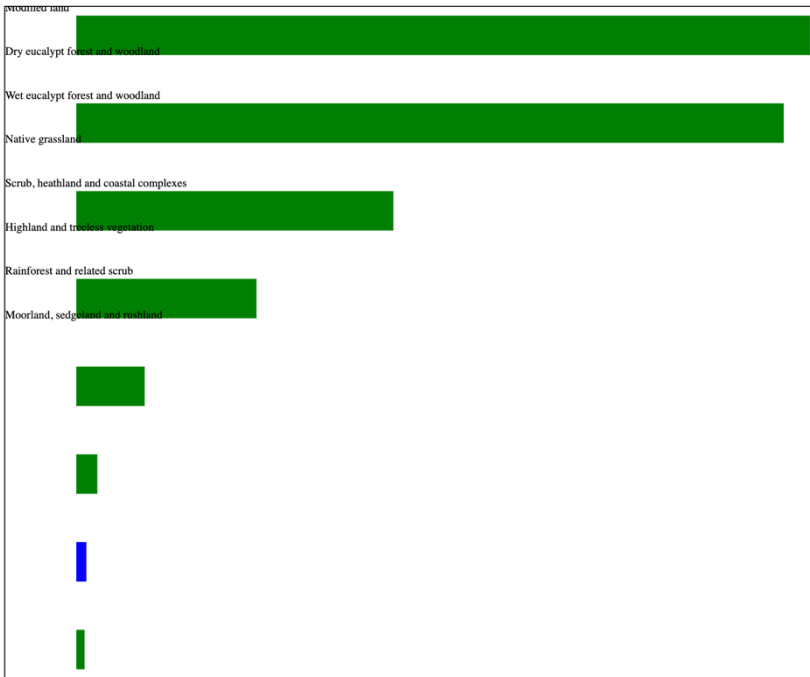
Adding category labels

barAndLabel

```
.append("text")
```

```
.text(d => d.habitat)
```

```
<svg viewBox="0 0 1200 1600" style="border: 1px solid black;">
  <g transform="translate(0, 6.1728395061728065)">
    <rect class="bar bar-1673" width="1032.7160493827162" height="55.555555555555556" x="100" y="6.1728395061728065"
      fill="green"></rect>
    <text>Modified land</text>
  </g>
```



Labelling

...

```
.attr("x", 250)
```

...

barAndLabel

```
.append("text")
```

```
.text(d => d.habitat)
```

```
.attr("x", 240)
```

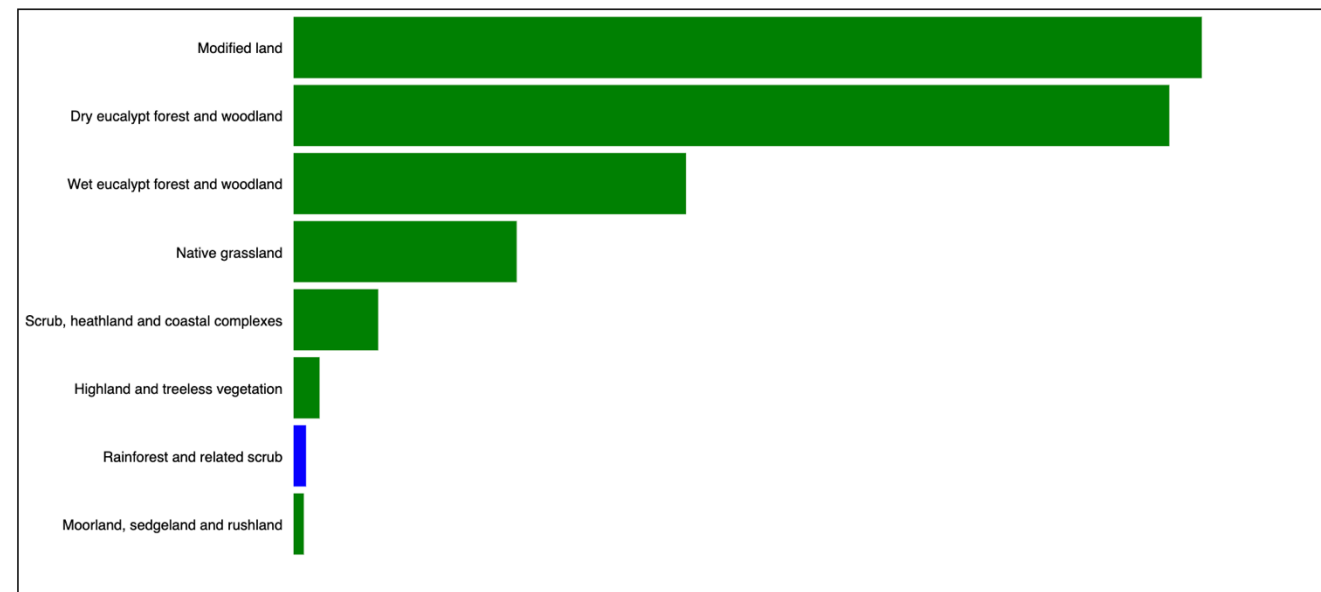
```
.attr("y", 33)
```

```
.attr("text-anchor", "end")
```

```
.style("font-family", "sans-serif")
```

```
.style("font-size", "13px");
```

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Adding value labels to chart

barAndLabel

```
.append("text")
.text(d => d.sighting)
.attr("x", d => 250 + xScale(d.sighting) + 4)
.attr("y", 33)
.style("font-family", "sans-serif")
.style("font-size", "13px");
```

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